Hence, the equation of normal is $(y-y_1) = \frac{(y_1+f)}{(x_1+g)}(x-x_1)$

$\Rightarrow \quad (y-y_1)(x_1+g) = (y_1+f)(x-x_1)$

$\Rightarrow \quad x_1(y-y_1)+g(y-y_1) = y_1(x-x_1)+f(x-x_1)$

$\Rightarrow \quad yx_1 - xy_1 + g(y-y_1) - f(x-x_1) = 0$.

Remark

(1) The equation of tangent at (x_1, y_1) to the circle $x^2+y^2=a^2$ is $xx_1 + yy_1 = a^2$.

(2) The equation of normal at (x_1, y_1) to the circle $x^2+y^2=a^2$ is $xy_1 - yx_1 = 0$.

(3) The normal passes through the centre of the circle.

5.2.3 Condition for the line $y = mx + c$ to be a tangent to the circle $x^2 + y^2 = a^2$ and finding the point of contact

Let the line $y = mx + c$ touch the circle $x^2 + y^2 = a^2$. The centre and radius of the circle $x^2 + y^2 = a^2$ are $(0,0)$ and a respectively.

(i) Condition for a line to be tangent

Then the perpendicular distance of the line $y - mx - c = 0$ from $(0,0)$ is

$$\left|\frac{0 - m.0 - c}{\sqrt{1+m^2}}\right| = \frac{|c|}{\sqrt{1+m^2}}.$$

This must be equal to radius .Therefore $\frac{|c|}{\sqrt{1+m^2}} = a$ or $c^2 = a^2(1+m^2)$.

Thus the condition for the line $y = mx + c$ to be a tangent to the circle $x^2 + y^2 = a^2$ is $c^2 = a^2(1+m^2)$.

(ii) Point of contact

Let (x_1, y_1) be the the point of contact of $y = mx + c$ with the circle $x^2 + y^2 = a^2$,

Then $y_1 = mx_1 + c$. ... (1)

Equation of tangent at (x_1, y_1) is $xx_1 + yy_1 = a^2$.

$$yy_1 = -xx_1 + a^2 \qquad \ldots (2)$$

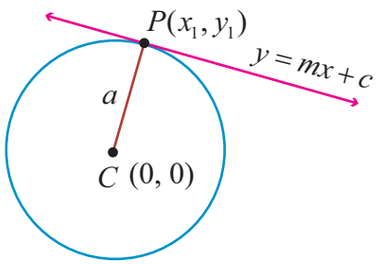


Fig.5.14

Equations (1) and (2) represent the same line and hence the coefficients are proportional.

So, $\frac{y_1}{1} = \frac{-x_1}{m} = \frac{a^2}{c}$

$$y_1 = \frac{a^2}{c}, x_1 = \frac{-a^2 m}{c}, \; c = \pm a\sqrt{1+m^2}.$$

Then the points of contact is either $\left(\frac{-am}{\sqrt{1+m^2}}, \frac{a}{\sqrt{1+m^2}}\right)$

or $\left(\frac{am}{\sqrt{1+m^2}}, \frac{-a}{\sqrt{1+m^2}}\right)$.

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